



Effective Power Management For Wearable Devices



A smart wearable whose battery runs out in a few hours is not desired by anyone. But by identifying the associated problems and carefully designing the power circuitry, this battery issue can be avoided

Vinay Prabhakar Minj

We are living in an era when electronic devices are an integral part of our daily life as these have made it very convenient for us to study, work, exercise, travel, communicate, and monitor things. Wearable devices, in particular, have become so lightweight and useful that many people like to carry them around every day.

In healthcare applications, wearables can be used to monitor heart-rate, blood pressure, SpO2 level, calories burnt during exercise, sleep tracking, and more. These devices monitor a patient's health, allow personal diagnosis, and fitness tracking.

Some of them also enable a wireless interface between a patient's doctor and the device for remote monitoring and control. This keeps the doctor informed about the patient's health status without either of them visiting the other.

For some neonatal cases, there are special forms of necklaces and even earbuds that can monitor a newborn child's body temperature and pulse. Such devices need a power management system that can last for at least nineteen hours before recharge.

Battery challenges

With so many features and functionalities, the battery management system (BMS) of a